HR Analytics

Employee Attrition Analysis

Prepared by:

Hassan Abd Alrazig Mohamed Hofney

Tools:

Python | Pandas | NumPy | Power BI



Project Overview

Project Goal

1-Analyze employee attrition.
2-Identify the main causes.
3-Support HR decision-making.

Dataset

Include Employee Information as :
Department-age-salary-jop Role-over time
Years at company- jop stasfaction
And other employee- related features

Data source

Data set : IBM HR Analytics Employee attription
And performance
source : Kaggle

Data clean in python

DistanceFromHome contained missing values (Null).

```
#destant from home is cleaned its was  value null
print(data["DistanceFromHome"])
print(data["DistanceFromHome"].isnull().sum())

data["DistanceFromHome"]=data["DistanceFromHome"].fillna(data["DistanceFromHome"].mean())
print(data["DistanceFromHome"].isnull().sum())
#data education is cleaned it was value eror int
```

Education contained invalid values.

```
#data education is cleaned it was value eror int
print(data["Education"].head())
print(data["Education"].value_counts())
print(data["Education"].isnull().sum())
n=0
for i in data["Education"]:
    if i=="1" or i=="2" or i=="3" or i =="4" or i =="5":
        n+=1
        continue
    else:
        data["Education"][n]="3"
        n+=1
print(data["Education"].value_counts())
print(data["EducationField"].head())
print(type(data["EducationField"][5]))
print(data.iloc[5])
```

EducationField contained invalid value.

```
c=[]
for i in data["EducationField"]:
    if i not in c:
        c.append(i)
    else:
        continue
print(c)
j=0
for i in data["EducationField"]:
    if i == "1" or i =="2" or i =="3" or i == "4" or i == "6":
        data["EducationField"][j]=="Other"
        j+=1
    else:
        j+=1
print(data["Education"].value_counts())
```

Data clean in python

EmployeeCount has the same value for all employees, so it has no variance

```
# data emplotee count is cleanend its dont need
print(data["EmployeeCount"].value_counts())
data=data.drop(columns=["EmployeeCount"])
```

EmployeeNumper was duplicated

```
#employee numper is cleaned it was duplicate
print(type(data["EmployeeNumber"][1]))
print(data["EmployeeNumber"].value_counts())

def function (s):
    c=[]
    v=0
    for i in s :
        if i in c :
            s.iloc[v]=np.random.randint(0,1000000000)
            v+=1
        else:
            c.append(i)
            v+=1
            continue
function(data["EmployeeNumber"])
print(data["EmployeeNumber"].value_counts())
#data Environmentsatisfaction is cleanend
```

Data clean in python

- Over18 column has the same value for all employees.
- The column was removed because it provides no useful information.

```
#data over 18 is cleanend it was not need it
data=data.drop(columns=["Over18"])
```

StandardHours column has the same value for all employees.

The column was removed because it has no variance.

```
#standarthour is cleaned no neaded
data=data.drop(columns=["StandardHours"])
############
```

Missing values were replaced with the mean value in Hourly Rate

```
#data HourlyRate is cleaned it was content value null
print(data["HourlyRate"])
print(data["HourlyRate"].value_counts())
print(data["HourlyRate"].isnull().sum())
x=0
data["HourlyRate"]=data["HourlyRate"].fillna(np.random.randint(0,100))
print(data["HourlyRate"].isnull().sum())
data.to_excel("hassan ad alrazig",index=False)
```

Recommendations

- **Provide competitive salaries for new employees**, as many employees leave the company due to low salaries.
- **Transfer employees who live far away to the nearest branch** whenever possible.
- **Reduce overtime working hours**, as a large number of employees resign because of excessive overtime.
- **Offer better salaries to software developers**, since most employees who leave the company are from this department due to salary concerns.